32-bit RISC-V processor

RTL implementation, functional verification and logic synthesis

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Five elastic stages: instruction fetch, decode, execute, memory and writeback. The physical design includes the processor and its iterative multiply/divide unit. Instruction/data memory, caches, the accelerator and chip IO pads are outside this GDS boundary.

Implementation platform: public Nangate45 / FreePDK45, typical Liberty corner at 1.1 V and 25 C. Clock target: 10 ns. Results are measured from saved tool runs; limitations are identified alongside the corresponding checks.

Prepared against the supplied assignment question paper. Screenshots are actual tool windows captured on an isolated Linux X11 display; terminal figures explicitly review recorded logs. Full source, logs and databases accompany the project.

Tools and reproducible environment

Tool / input	Version or identity
yosys	Yosys 0.9 (git sha1 1979e0b)
iverilog	Icarus Verilog version 11.0 (stable) ()
verilator	Verilator 4.038 2020-07-11 rev v4.036-114-g0cd4a57ad
openroad	v2.0-17598-ga008522d8
klayout	KLayout 0.30.12
gcc	riscv64-unknown-elf-gcc () 10.2.0

Platform checkout: 6101364b2d7909dd797e1e3e7f80695401cfa4e4. The official Ubuntu 22.04 OpenROAD release and its SHA256 are recorded in reports/toolchain_installation.json. No commercial libraries were required.

Toolchain verification

Actual version-command output captured in xterm. The installation package and platform revision are pinned in the project.

```
+ cat build/evidence/tools_review.txt
OPEN-SOURCE TOOLCHAIN - actual installed versions

yosys: Yosys 0.9 (git sha1 1979e0b)
iverilog: Icarus Verilog version 11.0 (stable) ()
verilator: Verilator 4.038 2020-07-11 rev v4.036-114-g0cd4a57ad
openroad: v2.0-17598-ga008522d8
klayout: KLayout 0.30.12
gcc: riscv64-unknown-elf-gcc () 10.2.0
platform_commit: 6101364b2d7909dd797e1e3e7f80695401cfa4e4

Platform: public Nangate45 / FreePDK45
Physical scope: core only; no SoC SRAM or pad ring
Analysis: typical corner; not foundry signoff
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Design hierarchy and stage responsibilities

Block	Implementation and purpose
rv32_pipeline	Physical top. Register file, decode/immediates, five valid-controlled stages, bypassing, memory handshakes, redirect/exception control, counters and machine CSRs.
rv32_muldiv	Iterative RV32M arithmetic unit. Blocks EX until the result is ready; signed and unsigned multiply/divide/remainder cases.
pipeline_evidence_tb	Directed test with an independent architectural scoreboard and explicit coverage counters.
cpu_tb and isa_reference.py	Randomized request/response latency; retirement trace checked against an independent instruction interpreter.
pipelined_soc and component RTL	Separately verified cache/fabric/accelerator extension; excluded from the core physical implementation.

This retains the proposed processor idea while using the existing RV32IM implementation. The fetch frontend has one outstanding request and does not sustain one instruction per cycle. No pipeline speedup or ISA certification is inferred from the five-stage structure.

Five-stage data and control flow



Stage registers carry PC, instruction, operand values and control state. A valid/ready-style elastic transfer allows MEM or EX to wait without losing instruction state. Stores use byte strobes; loads shift and extend the selected byte or halfword.

Bypasses select the newest matching result from the EX/MEM and MEM/WB paths. A dependent instruction waits while a load result is unavailable. Taken branches, JAL/JALR and exceptions cancel younger instructions and redirect fetch; pending wrong-path responses are marked killed.

The register file forces x0 to zero. CSR and fence operations serialize as needed. The machine-mode environment includes directed exception and interrupt support, but does not implement an MMU, user/supervisor modes or a full privileged architecture.

RTL control source

Source review of the load-use interlock, redirect condition and data interface assignments. Complete modules are supplied under rtl/.

```
+ cat build/evidence/rtl_review.txt
RTL IMPLEMENTATION - source review
Physical top: rv32_pipeline + rv32_muldiv
Five elastic stages; instruction and data handshake ports
RTL sources: rtl/rv32_pipeline.sv, rtl/rv32_muldiv.sv

   if(taken&&next_pc[1:0]!=0)begin exception=1;cause=0;tval=next_pc;end
   if(illegal)begin exception=1;cause=2;tval=ex_insn;end
   if(ex_error)begin exception=1;cause=1;tval=ex_pc;end
   if(exception)begin regwrite=0;isload=0;isstore=0;csrwrite=0;end
 end
 wire ex_redirect=ex_go&&(taken||exception);
 wire ex_load_hazard=ex_valid&&isload&&ex_insn[11:7]!=0&&
  (ex_insn[11:7]==id_insn[19:15]||ex_insn[11:7]==id_insn[24:20]);
 wire id_go=id_valid&&(!ex_valid||ex_go)&&!ex_load_hazard&&!ex_redirect&&!(mem_done&&mem_fault);
 wire fetch_accept=iv&&ir;
 assign iv=!rst&&fetch_queued;
 assign ia=fetch_addr;
 assign dv=mem_valid&&(mem_load||mem_write)&&!mem_sent&&!mem_trap;
 assign da=mem_addr;assign dw=mem_write;assign wd=mem_store;assign ws=mem_write?mem_strobe:0;
 integer n;
 task enter_trap(input [31:0] pc_value,input [31:0] cause_value,input [31:0] tval_value);
 begin
  mepc<=pc_value;mcause<=cause_value;mtval<=tval_value;
  mstatus[7]<=mstatus[3];mstatus[3]<=0;mstatus[12:11]<=3;
  fetch_pc<={mtvec[31:2],2'b00}+((mtvec[1:0]==1&&cause_value[31])?4*{1'b0,cause_value[30:0]}:0);
  id_valid<=0;ex_valid<=0;mem_valid<=0;fetch_killed<=(fetch_pending||fetch_queued)&&!ip;
  trap_valid<=1;trap_cause<=cause_value;
 end endtask
 always @(posedge clk) begin
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Forwarding and load-use control

Exact source excerpts from rv32_pipeline.sv. The newest valid result wins bypass priority; the interlock prevents a dependent ID instruction from consuming an unavailable load result.

```
  (wb_valid&&wb_regwrite&&wb_rd!=0&&wb_rd==ex_rs1)?wb_value:ex_a;
wire [31:0] b=(mem_valid&&mem_regwrite&&!mem_fault&&mem_rd!=0&&mem_rd==ex_rs2&&(!mem_load||mem_done))?mem_value:
  (wb_valid&&wb_regwrite&&wb_rd!=0&&wb_rd==ex_rs2)?wb_value:ex_b;
reg [31:0] result,address,store_data,next_pc,tval,csr_old,csr_new;
reg [31:0] cause;
reg regwrite,isload,isstore,illegal,taken,exception,csrwrite,is_mret,is_fencei,csr_ok;
reg [3:0] strobes;

integer n;
task enter_trap(input [31:0] pc_value,input [31:0] cause_value,input [31:0] tval_value);
begin
 mepc<=pc_value;mcause<=cause_value;mtval<=tval_value;
 mstatus[7]<=mstatus[3];mstatus[3]<=0;mstatus[12:11]<=3;
 fetch_pc<={mtvec[31:2],2'b00}+((mtvec[1:0]==1&&cause_value[31])?4*{1'b0,cause_value[30:0]}:0);
 id_valid<=0;ex_valid<=0;mem_valid<=0;fetch_killed<=(fetch_pending||fetch_queued)&&!ip;
 trap_valid<=1;trap_cause<=cause_value;
end endtask
```

Complete rv32_pipeline.sv, rv32_muldiv.sv and pipeline_evidence_tb.sv are embedded as file attachments in this PDF and supplied as normal project files.

Stage transfer and architectural writeback

Exact stage-transfer source: ID captures operand values with bypassing, and WB exposes retirement and updates the architectural register file. Valid bits govern which payloads are meaningful.

```
if(id_go)begin
 id_valid<=0;ex_valid<=1;ex_pc<=id_pc;ex_insn<=id_insn;ex_error<=id_error;
 ex_a<=id_insn[19:15]==0?0:
   (ex_go&&regwrite&&ex_insn[11:7]==id_insn[19:15])?result:
   (mem_valid&&mem_regwrite&&!mem_fault&&mem_rd==id_insn[19:15]&&(!mem_load||mem_done))?mem_value:
   (wb_valid&&wb_regwrite&&wb_rd==id_insn[19:15])?wb_value:regs[id_insn[19:15]];
 ex_b<=id_insn[24:20]==0?0:
   (ex_go&&regwrite&&ex_insn[11:7]==id_insn[24:20])?result:
   (mem_valid&&mem_regwrite&&!mem_fault&&mem_rd==id_insn[24:20]&&(!mem_load||mem_done))?mem_value:
   (wb_valid&&wb_regwrite&&wb_rd==id_insn[24:20])?wb_value:regs[id_insn[24:20]];
end
if(dv&&dr)mem_sent<=1;
if(wb_valid)begin
 retire_valid<=1;retire_pc<=wb_pc;retire_insn<=wb_insn;
 retire_rd<=wb_regwrite?wb_rd:0;retire_value<=wb_value;instret<=instret+1;
 if(wb_regwrite&&wb_rd!=0)regs[wb_rd]<=wb_value;
 if(wb_csrwrite)case(wb_csraddr)
  12'h300:mstatus<=(wb_csrvalue&32'h88)|32'h1800;
  12'h304:mie<=wb_csrvalue&32'h880;
  12'h305:mtvec<={wb_csrvalue[31:2],1'b0,wb_csrvalue[0]};12'h340:mscratch<=wb_csrvalue;
  12'h341:mepc<={wb_csrvalue[31:2],2'd0};12'h342:mcause<=wb_csrvalue;12'h343:mtval<=wb_csrvalue;
  12'hb00:cycles[31:0]<=wb_csrvalue;12'hb80:cycles[63:32]<=wb_csrvalue;
  12'hb02:instret[31:0]<=wb_csrvalue;12'hb82:instret[63:32]<=wb_csrvalue;
  default:begin end
```

Verification plan and measured coverage

Requirement / test	Measured evidence
Forwarding and load-use stall	Directed test: forwarding=2; load-use stall=13 cycles; architectural scoreboard passes.
Control hazards	Two redirects; PC 0x20 and 0x28 are forbidden from retirement; taken and untaken branches plus JAL checked.
Architectural state	Expected x1=7, x2=12, x3=12, x4=24, x6=0, x7=x8=40. Test finishes at cycle 70 with 12 retirements observed before the exit-store response.
RV32M arithmetic	520 multiply/divide test vectors including edge cases.
CPU differential verification	8176 retirements at delay 0; 8176 retirements at delay 3; 8176 retirements at delay 7.
Architectural suite	49 passing programs; adapted riscv-tests environment, not certification.
Additional regression	Cache, AXI adapter, traps, matrix/DMA and integrated firmware sweeps; fatal lint and a 25-cycle bounded arbiter safety proof.

Directed pipeline test results

The recorded run prints each retirement and the scoreboard/coverage result. Wrong-path instructions are rejected by PC; the exit store completes after the reported 12 retirements.

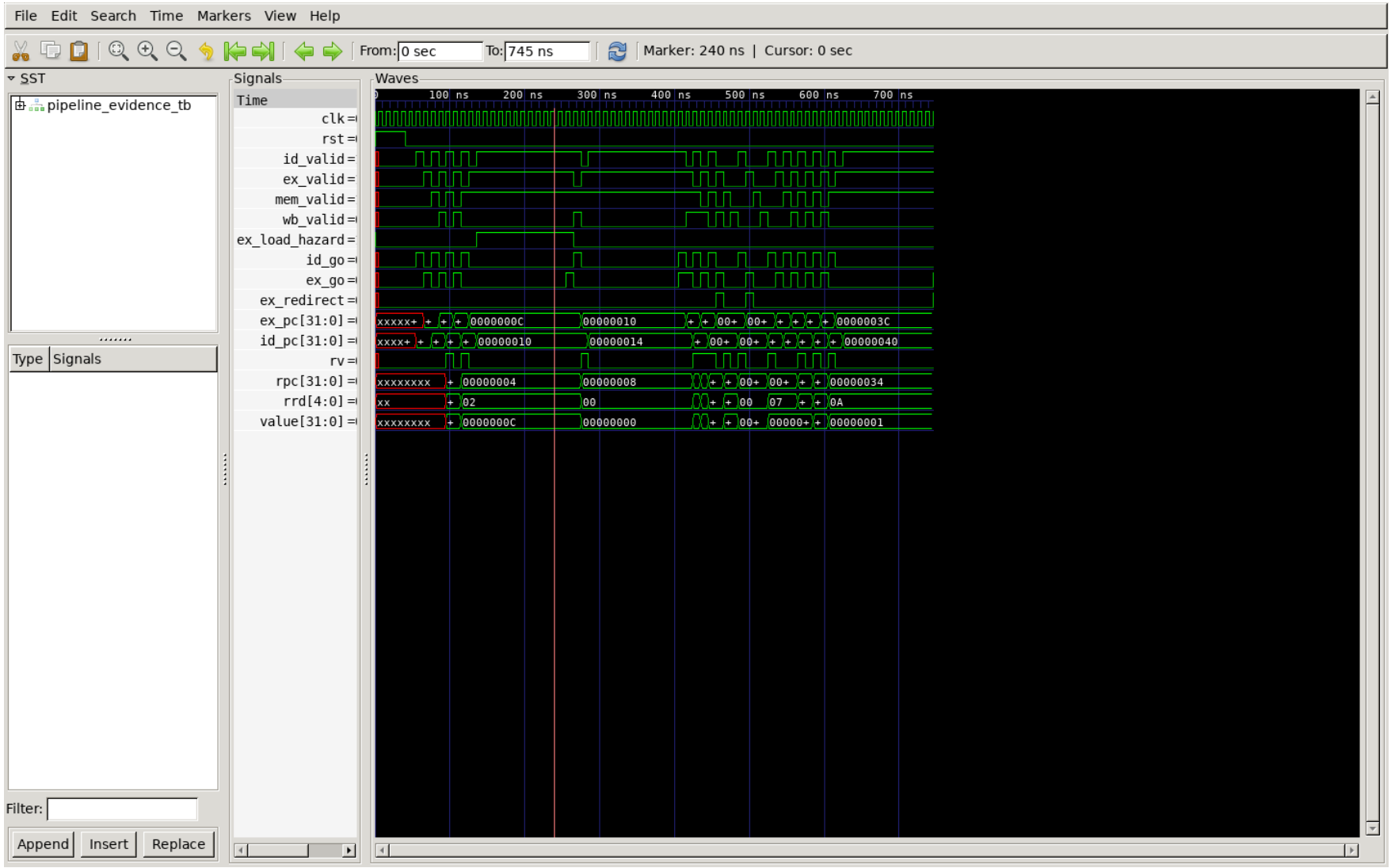
```
+ cat build/evidence/verification_review.txt
FUNCTIONAL VERIFICATION - recorded run output
Command: python3 scripts/pipeline_evidence.py
Source log: build/evidence/hazards.log

VCD info: dumpfile build/evidence/pipeline.vcd opened for output.
RETIRE pc=00000000 insn=00700093 rd=1 value=00000007
RETIRE pc=00000004 insn=00508113 rd=2 value=0000000c
RETIRE pc=00000008 insn=10202023 rd=0 value=00000000
RETIRE pc=0000000c insn=10002183 rd=3 value=0000000c
RETIRE pc=00000010 insn=00218233 rd=4 value=00000018
RETIRE pc=00000014 insn=00020463 rd=0 value=00000000
RETIRE pc=00000018 insn=01800293 rd=5 value=00000018
RETIRE pc=0000001c insn=00520463 rd=0 value=00000000
RETIRE pc=00000024 insn=008003ef rd=7 value=00000028
RETIRE pc=0000002c insn=00038413 rd=8 value=00000028
RETIRE pc=00000030 insn=400004b7 rd=9 value=40000000
RETIRE pc=00000034 insn=00100513 rd=10 value=00000001
PASS directed architectural scoreboard; wrong-path PCs excluded
COVER forwarding=2 load_use_stall_cycles=13 redirects=2
MEASURE cycles=70 retired=12
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Pipeline waveform in GTKWave

The asserted load-hazard interval holds ID while memory is busy. ex_redirect pulses correspond to the taken branch and jump. X values before a stage becomes valid are uninitialized, invalid payloads.



Full regression and reference comparisons

Saved make all output and generated JSON results. Whole-SoC checks extend beyond the processor core that is physically implemented.

```
+ cat build/evidence/regression_review.txt
FULL REGRESSION - saved log and JSON artifacts
Command: make all
Source: build/evidence/regression.log

Multiply/divide vectors: 520
CPU delay=0: compared 8176 retirements
CPU delay=3: compared 8176 retirements
CPU delay=7: compared 8176 retirements
Architectural result file: reports/architectural_tests.json

PASS KERNEL pipeline=1 delay=0 clocks=562536 kernel_cycles=371361 hash=2e60bc73
WARNING: tb/kernel_tb.sv:26: $readmemh(build/kernel_1.hex): Not enough words in the file for the requested ran
535].
PASS KERNEL pipeline=1 delay=5 clocks=1289779 kernel_cycles=850609 hash=2e60bc73
00000154 B _end
00000000 T _start
00000144 B kernel_metrics
00000018 T main
WARNING: tb/kernel_tb.sv:26: $readmemh(build/kernel_2.hex): Not enough words in the file for the requested ran
535].
PASS KERNEL pipeline=1 delay=0 clocks=70817 kernel_cycles=57138 hash=2e60bc73
WARNING: tb/kernel_tb.sv:26: $readmemh(build/kernel_2.hex): Not enough words in the file for the requested ran
535].
PASS KERNEL pipeline=1 delay=5 clocks=116487 kernel_cycles=94397 hash=2e60bc73
python3 models/dataflows.py
PASS dataflow models and independent inference: [-1, -1, 0]
python3 scripts/report.py
PASS generated reports from measured regression artifacts
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Technology synthesis flow and constraints

Yosys elaborates and flattens rv32_pipeline and rv32_muldiv, lowers processes and memories, maps flip-flops with dfflibmap, then maps combinational logic through ABC using the Nangate45 typical Liberty library. Tie cells implement constants. Structural checks run before and after mapping.

```
read_verilog -sv rtl/rv32_pipeline.sv rtl/rv32_muldiv.sv
synth -top rv32_pipeline -flatten
dfflibmap -liberty NangateOpenCellLibrary_typical.lib
abc -liberty NangateOpenCellLibrary_typical.lib -D 10000
hilomap -hicell LOGIC1_X1 Z -locell LOGIC0_X1 Z
check -assert
write_verilog -noattr -noexpr build/asic/1_synth.v
```

The 10 ns clock is a target. IO budgets are 1 ns; output load is 0.01 pF and input transition 0.1 ns. Setup uncertainty is 0.2 ns and hold uncertainty 0.05 ns. The register file maps to standard cells; no instruction/data SRAM is part of synthesis.

Synthesis results and interpretation

Metric	Measured result
Mapped cells	17,853 cells including tie cells
Mapped cell area	29,685.866 square micrometres
Clock / corner	10 ns; Nangate45 typical, 1.1 V / 25 C
Initial setup slack	-5.64 ns; ideal clock, no routed interconnect
Initial hold slack	0.03 ns under the revised 0.05 ns hold uncertainty
Inference boundary	Core register file and arithmetic are mapped; external memory and optional SoC modules are excluded.

Initial synthesis timing does not meet the setup target and includes electrical violations from the minimally driven mapped netlist. Physical placement and repair resize cells and insert buffers; the later back-end report presents those measured improvements. Initial synthesis slack is not a post-route frequency result.

Mapped Yosys synthesis output

Recorded standard-cell statistics and Liberty area. The area excludes external instruction/data memories, caches, accelerator and IO pads.

```
+ cat build/evidence/synthesis_review.txt
TECHNOLOGY SYNTHESIS - recorded Yosys output
Command: python3 asic/open/synth.py
Library: NangateOpenCellLibrary_typical.lib
Source log: build/asic/1_synth.log

     MUX2_X1                     1688
     NAND2_X1                    1275
     NAND3_X1                    1041
     NAND4_X1                      44
     NOR2_X1                     1443
     NOR3_X1                      242
     NOR4_X1                       47
     OAI211_X1                    421
     OAI21_X1                    2390
     OAI221_X1                    204
     OAI222_X1                     16
     OAI22_X1                     272
     OAI33_X1                       5
     OR2_X1                       296
     OR3_X1                        84
     OR4_X1                        13
     XNOR2_X1                     470
     XOR2_X1                      325

   Chip area for module '\rv32_pipeline': 29685.866000

12. Executing Verilog backend.
Dumping module `\rv32_pipeline'.

13. Executing JSON backend.
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Initial timing paths in OpenROAD

The mapped netlist is loaded into the OpenROAD Timing Report. No placement geometry is expected at this stage; timing is based on ideal interconnect. The displayed 100% is an unplaced-tool default, not physical core utilization.

Timing Report

Settings | Columns | Update

Setup | Hold

Capture Clock	Required	Arrival	Slack	Skew	Logic Delay	Logic Depth	Fanout	Start	End
core_clk	9.762	15.399	-5.636	0.200	14.399	6	1837	rst	_34217_/D
core_clk	9.762	15.393	-5.631	0.200	14.393	6	1837	rst	_34256_/D
core_clk	9.763	15.393	-5.630	0.200	14.393	5	1836	rst	_34414_/D
core_clk	9.764	15.390	-5.626	0.200	14.390	6	1837	rst	_34191_/D
core_clk	9.764	15.390	-5.626	0.200	14.390	6	1837	rst	_34193_/D
core_clk	9.764	15.390	-5.626	0.200	14.390	6	1837	rst	_34209_/D
core_clk	9.764	15.390	-5.626	0.200	14.390	6	1837	rst	_34216_/D
core_clk	9.767	15.393	-5.626	0.200	14.393	8	1839	rst	_34350_/D
core_clk	9.767	15.393	-5.626	0.200	14.393	8	1839	rst	_34351_/D
core_clk	9.767	15.393	-5.626	0.200	14.393	8	1839	rst	_34354_/D
core_clk	9.767	15.393	-5.626	0.200	14.393	8	1839	rst	_34356_/D
core_clk	9.767	15.393	-5.626	0.200	14.393	8	1839	rst	_34359_/D
core_clk	9.767	15.393	-5.626	0.200	14.393	8	1839	rst	_34360_/D
core_clk	9.767	15.393	-5.626	0.200	14.393	8	1839	rst	_34362_/D

Data Path Details | Capture Path Details

Pin	Fanout	⇅	Time	Delay	Slew	Load
rst	352	↓	1.000	1.000	0.100	
clock network delay			1.000	0.000		
_16598_/A (INV_X1)		↓	1.000	0.000	0.100	
_16598_/ZN (INV_X1)	1111	↑	5.298	4.298	3.821	1791.892
_17301_/A1 (NAND2_X1)		↑	5.298	0.000	3.821	
_17301_/ZN (NAND2_X1)	222	↓	13.386	8.088	0.379	324.012
_29018_/A1 (NOR2_X1)		↓	13.386	0.000	0.379	
_29018_/ZN (NOR2_X1)	150	↑	15.236	1.850	1.123	240.710
_29996_/A (OAI211_X1)		↑	15.236	0.000	1.123	
_29996_/ZN (OAI211_X1)	1	↓	15.332	0.096	0.157	1.498
_29997_/B (OAI211_X1)		↓	15.332	0.000	0.157	
_29997_/ZN (OAI211_X1)	1	↑	15.399	0.067	0.041	1.140
_34217_/D (DFF_X1)		↑	15.399	0.000	0.041	

Mapped and routed logical regression

The same independent directed architectural scoreboard passes on the routed, mapped netlist using Liberty-derived functional cell models. This is zero-delay simulation without SDF, and is separate from physical LVS.

```
+ cat build/evidence/gate_review.txt
ZERO-DELAY MAPPED NETLIST CHECK
Command: python3 asic/open/gate_check.py
Source: build/evidence/7_final_gate.log

VCD info: dumpfile build/evidence/7_final_gate.vcd opened for output.
RETIRE pc=00000000 insn=00700093 rd=1 value=00000007
RETIRE pc=00000004 insn=00508113 rd=2 value=0000000c
RETIRE pc=00000008 insn=10202023 rd=0 value=00000000
RETIRE pc=0000000c insn=10002183 rd=3 value=0000000c
RETIRE pc=00000010 insn=00218233 rd=4 value=00000018
RETIRE pc=00000014 insn=00020463 rd=0 value=00000000
RETIRE pc=00000018 insn=01800293 rd=5 value=00000018
RETIRE pc=0000001c insn=00520463 rd=0 value=00000000
RETIRE pc=00000024 insn=008003ef rd=7 value=00000028
RETIRE pc=0000002c insn=00038413 rd=8 value=00000028
RETIRE pc=00000030 insn=400004b7 rd=9 value=40000000
RETIRE pc=00000034 insn=00100513 rd=10 value=00000001
PASS directed architectural scoreboard; wrong-path PCs excluded
MEASURE cycles=70 retired=12
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Discussion of front-end results

The directed evidence demonstrates the required dependency and redirect behavior. The broader regression checks instruction semantics and several memory latency configurations. The observed CPI reflects the conservative fetch protocol and memory stalls; five stages alone do not imply ideal throughput.

Technology mapping succeeded, but the initial setup slack is negative. Placement-time resizing substantially changes both timing and area. The front-end report therefore retains the synthesis result and leaves the physically optimized result to Assignment X.

The existing project includes more functionality than the initial RV32I proposal: iterative RV32M, a limited machine-mode environment and optional SoC modules. Only the processor top and M unit are included in the GDS. Formal evidence covers the arbiter for 25 bounded cycles; there is no whole-CPU formal proof or complete privileged-ISA certification.

A short readmemh image warning in the older regression means the initialized simulation memory is larger than the loaded program. The directed evidence image uses an explicit range. Assertions and architectural comparisons, rather than the absence of warning text, establish the reported pass results.

Contributions, scope and reproduction

Contributor / assistance	Recorded role
Shiv Sriram V, 24BVD0168	Project owner; supplied the processor proposal and existing project repository. No additional team members were provided.
Existing repository	Processor, optional SoC extensions, firmware and regression infrastructure predated this session.
AI-assisted work in this session	Directed hazard test and waveform; technology-mapped synthesis; mapped/post-route directed netlist checks; tool execution, evidence capture and report compilation. Student review and approval of the submitted claims remain with the project owner.

No commits were made or staged. Suggested stage-by-stage commit messages and file groupings are in docs/COMMIT_GUIDE.md. The project includes exact constraints, source scripts, tool outputs and screenshots for independent reproduction.

```
make pipeline
make all
make asic
make physical-checks
make evidence
make submissions
```

Source and evidence index

Assignment sources: the supplied Assignment VIII project proposal; ASIC Design Lab QP 9 (1).pdf (Assignment IX, Front End); ASIC Design Lab QP 0.pdf (Assignment X, Back End). Submission 8 duplicates the supplied proposal.

OpenROAD flow and public platform documentation:
https://openroad-flow-scripts.readthedocs.io/en/latest/
Official binary release:
https://github.com/Precision-Innovations/OpenROAD/releases/tag/2024-12-14

Evidence location	Content
rtl/ and tb/	Synthesizable sources and independent directed scoreboard
build/evidence/	Full regression, hazard/gate logs and waveforms; terminal review text
build/asic/	Synthesis log/netlist/JSON; ODB, DEF, SDC, STA paths, SPEF, GDS and DRC database
reports/asic_metrics.json	Measured area, utilization, setup/hold slack and tool/platform versions
reports/physical_checks.json	Independent GDS DRC outcome and explicit LVS availability
reports/evidence/	Unaltered actual GUI and xterm screenshot PNG files
asic/open/README.md	Reproduction, constraints, scope and tool setup